KENYA JUNIOR SCHOOL EDUCATION ASSESSMENT

GRADE 9 TERM 1 MARCH 2026

INTEGRATED SCIENCE CKEAB 003

LEARNER'S DETAILS

Name: _____

School: _____

Assessment Number: _____. **Date:** _____

School Code: _____. **Signature:** _____

INSTRUCTIONS TO LEARNERS

1. Write your name and class on the answer sheet provided.
2. Answer **ALL questions** in **Section A** and **Section B**.
3. For questions with pictures, study the pictures carefully before answering.
4. Ensure your work is neat and clear.

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Question Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Maximum Score	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Candidate's Score															

Question Number	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	Total score
Maximum Score	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	30
Candidate's Score																

SECTION B: STRUCTURED QUESTIONS. (50 Marks)

Qn No.	31	32	33	34	35	36	Total
Max Score	5	8	6	6	10	15	**50**
Cand score							

Section	Maximum Marks	Learner's Score	Percentage (%)
A. Multiple Choice	30		
B. Structured Questions	50		
TOTAL	**80**		

Performance Levels Grid

Level	Level	% Range	Points	Learner's score	Remarks
Exceeding (EE)	EE1	90–100%	8		Exceptional mastery; exceeds expectations
	EE2	75–89%	7		Very strong performance
Meeting (ME)	ME1	58–74%	6		Solid understanding
	ME2	41–57%	5		Partial achievement
Approaching (AE)	AE1	31–40%	4		Limited achievement
	AE2	21–30%	3		Very limited
Below (BE)	BE1	11–20%	2		Poor performance
	BE2	1–10%	1		Extremely weak

-INTEGRATED SCIENCE- -003- - Paper Code 90

This paper has 6 printed pages. Please check out to ensure all pages are printed

Turn over

SECTION A: (30 MARKS)

Answer all the questions. Choose the correct answer for each.

1. Grade 9 learners were revising renewable and non-renewable sources of energy. Which one is a **non-renewable** source?
A. Wind B. Solar C. Coal D. Biogas

2. Photosynthesis is the process by which green plants make their own food. Which gas is mainly released?
A. Carbon dioxide B. Oxygen
C. Nitrogen D. Methane

3. Which statement describes **Osmosis**?
A. Water moves from low to high concentration
B. Water moves from high to low concentration through a semi-permeable membrane
C. Solutes move from high to low concentration
D. Gases move through a semi-permeable membrane

4. The smallest unit of an element that retains its chemical properties is:
A. Molecule B. Compound
C. Atom D. Ion

5. Grade 9 learners were discussing alloys. Which learner gave an **incorrect** statement?

Melvin: Alloys are mixtures of chemical elements creating metal-like properties.
Ochieng: Elements are added to metals to increase corrosion resistance.
Alice: Alloys are formed by combining two or more elements to create new materials.
Joyce: Alloys are both non-metals

A. Joyce B. Ochieng C. Alice D. Melvin

6. The diagram shows a plant cell under a microscope.

Which part represents a **chloroplast**?
A. R
B. O
C. Q
D. S

7. The part marked **O** in the diagram:
A. Carries genetic information and controls cell activities
B. Is a jelly-like liquid inside the cell
C. Acts as a fence and controls movement in/out of the cell
D. Absorbs light for photosynthesis

8. Which factor does **not** affect the rate of osmosis?
A. Temperature
B. Concentration gradient
C. Type of semi-permeable membrane
D. Container's surface area

9. Adolescents experience challenges during their menstrual cycle. Which one is **not** a challenge?
A. Irregular periods
B. Urge for sexual intercourse
C. Painful periods
D. Irregular bleeding

10. When a sperm penetrates an ovum, the fusion of their nuclei forms:
A. Foetus
B. Embryo
C. Zygote
D. Baby

11. Which element is correctly matched with its chemical symbol?
A. Sodium – So
B. Copper – Cu
C. Iron – Ag
D. Silver – Si

12. Grade 9 learners heated an iron over a flame until it became red hot. Which statement is **not true**?
A. The iron became red hot because it is a good conductor of heat
B. The change was a temporary change
C. The change was a permanent change
D. The change was physical

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13. Which of the following is **not a chemical change**?
A. Water boiling
B. Ripening of a fruit
C. Fermentation of grapes
D. Digestion of food

14. The largest component of blood is:
A. Red blood cells
B. White blood cells
C. Platelets
D. Plasma

15. Which energy transformation occurs in a flashlight?
A. Kinetic → Electrical → Light
B. Electrical → Chemical → Light
C. Chemical → Light → Kinetic
D. Chemical → Electrical → Light

16. During a science lesson, learners studied the structure of an atom using a model. Which particles are found in the **nucleus** of an atom?
A. Electrons only
B. Protons and neutrons
C. Protons and electrons
D. Neutrons and electrons

17. A learner rubbed a plastic ruler on dry hair and it attracted small pieces of paper. Which sub-atomic particle was transferred during rubbing?
A. Proton
B. Neutron
C. Electron
D. Nucleus

18. An element has an electron arrangement of **2, 8, 1**. Which statement about the element is correct?
A. It is a noble gas
B. It is likely to lose one electron
C. It is chemically unreactive
D. It has a full outer shell

19. During a laboratory activity, learners classified substances as elements, mixtures, or compounds. Which of the following is an **element**?
A. Water
B. Salt
C. Oxygen
D. Air

20. A mixture of **sand and iron filings** was prepared in class. Which method is most suitable for separating the mixture?
A. Filtration
B. Evaporation
C. Magnetic separation
D. Distillation

21. When **sodium reacts with chlorine** to form sodium chloride, the product formed is best described as a:
A. Mixture
B. Element
C. Compound
D. Solution

22. A metal spoon and a plastic spoon were placed in hot tea. The metal spoon became hot faster. Which property of matter is demonstrated?
A. Density
B. Conductivity
C. Elasticity
D. Solubility

23. Which of the following is a **physical change**?
A. Burning of charcoal
B. Rusting of iron
C. Melting of ice
D. Digestion of food

24. A candle was lit in a dark room. Which change observed is **permanent**?
A. Melting of wax
B. Solidifying of wax
C. Burning of wax
D. Cooling of wax

25. Cooking oil caught fire in a school kitchen. Which **class of fire** was involved?
A. Class A
B. Class B
C. Class C
D. Class D

26. A microscope has an eyepiece lens of **×10** and an objective lens of **×40**. What is the total magnification?
A. ×4
B. ×50
C. ×400
D. ×4000

27. A bottle of perfume was opened in a classroom, and after some time the smell spread throughout the room. This process is called:
A. Osmosis
B. Diffusion

C. Evaporation
D. Condensation

28. Which of the following factors affects the **rate of osmosis**?
A. Shape of the container
B. Temperature
C. Colour of the solution
D. Size of the container

29. Fertilization in human beings normally takes place in the:
A. Ovary B. Uterus
C. Oviduct D. Vagina

30. A learner applied a force of **100 N** on an area of **2 m²** while pushing a box. What pressure was exerted?
A. 20 Pa
B. 50 Pa
C. 100 Pa
D. 200 Pa

SECTION B: (40 MARKS)

Answer all the questions. Choose the correct answer for each.

31. Draw and label the structure of an atom, showing **protons, neutrons, and electrons**. (3 marks)

32. Give **three common methods** used to separate **solid-solid mixtures**. (3 marks)

a. _____.
b. _____.
c. _____.

33. (a)Describe the **total magnification of a microscope** and explain how it is determined. (3 marks)

a. _____.
b. _____.
c. _____.

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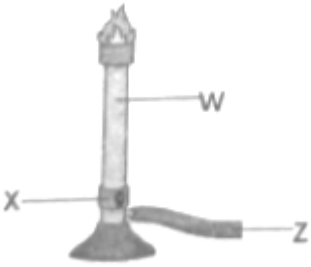
(b). The diagram below shows different meaning related to safety. Identify each of them (3mks)

_____. _____. _____.

34. Magnets can move objects without touching them. This invisible force is called: _____ (2 marks)

35. The diagram below shows a **Bunsen burner**. Identify the parts:



(a) X: _____.

(b) W: _____.

36. Name the laboratory equipment used for the following:

(a) Holding solid substances: _____.

(b) Transferring small amounts of liquid: _____.

37. Distinguish between **pure** and **impure** substances based on melting and boiling points. (2 marks)

_____

_____

_____

38. The diagram shows **safety signs**. Identify each:

(i) _____

(ii) _____

(iii) _____ (3 marks)

39. A patient with swollen legs was diagnosed with kidney disorder. List **four causes** of kidney disorders. (4 marks)

a. _____.

b. _____.

c. _____.

d. _____.

40. Complete the table by filling in the **colors observed when indicators** are put into solutions. (8 marks)

Indicator	Colour in phenolphalein	Colour in litmus paper	Colour in methyl orange
Wood ash solution			
Lemon juice solution			
Anti-acid table solution			
Tooth paste solution			

41. Identify the **elements and compounds** used in making the following: (3 marks)

(i) Toothpaste: _____.

(ii) Soaps and detergents: _____.

(iii) Tablet salt: _____.

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MARKING GUIDE

SECTION A:

1. **C**
2. **B**
3. **B**
4. **C**
5. **A**
6. **D**
7. **B**
8. **D**
9. **B**
10. **C**
11. **B**
12. **C**
13. **A**
14. **D**
15. **D**
16. **B**
17. **C**
18. **B**
19. **C**
20. **C**
21. **C**
22. **B**
23. **C**
24. **C**
25. **B**
26. **C**
27. **B**
28. **B**
29. **C**
30. **B**

SECTION B: ANSWERS (31–41)

31. Structure of an atom (3 marks)

- **Nucleus** – contains **protons** and **neutrons**
- **Electrons** – found in shells/energy levels around the nucleus
  (Award marks for correct labels of proton, neutron, electron)

32. Methods used to separate solid–solid mixtures (3 marks)

a) **Magnetic separation**
b) **Sieving**
c) **Hand picking**

33. Total magnification of a microscope (3 marks)

a) Total magnification is the **overall enlargement** of an object seen under a microscope.
b) It is determined by **multiplying the eyepiece magnification by the objective magnification**.
c) **Formula:**
Total magnification = Eyepiece × Objective

(Safety symbols – 3 marks)

33(b)(Any correct identification earns marks)

i. **Fire assembly point**
ii. **Fire exit**
iii. **Fire alarm**

34. Magnetic force (2 marks)

The invisible force is called **magnetism** or **magnetic force**.

35. Parts of a Bunsen burner (2 marks)

a) **X – Collar**
b) **W –Chimney (barrel)**

36. Laboratory equipment (2 marks)

a) Holding solid substances: **Spatula**
b) Transferring small amounts of liquid: **Dropper / Pipette**

37. Pure vs impure substances (2 marks)

- **Pure substances** have **fixed melting and boiling points**.
- **Impure substances** melt and boil over a **range of temperatures**.

38. Safety signs (3 marks)

(i) **Flammable**
(ii) **Corrosive**
(iii) **Toxic / Poisonous**

39. Causes of kidney disorders (4 marks)

a) **Excessive alcohol consumption**
b) **Untreated infections**
c) **High blood pressure**
d) **Diabetes**

40. Indicator colour changes (8 marks)

Solution	Phenolphthalein	Litmus Paper	Methyl Orange
Wood ash solution (alkaline)	Pink	Blue	Yellow
Lemon juice solution (acidic)	Colourless	Red	Red
Antacid tablet solution (alkaline)	Pink	Blue	Yellow
Toothpaste solution (alkaline)	Pink	Blue	Yellow

41. Elements/compounds used (3 marks)

i) Toothpaste: **Fluoride / Calcium carbonate**
ii) Soaps and detergents: **Sodium hydroxide / Sodium salts**
iii) Table salt: **Sodium chloride**

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